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# A two-plaquette polaron picture for the single-hole state of cuprates

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## Abstract

The pseudogap phase in high  $T_c$  superconductors has attracted intense research efforts in recent years. This phase is expected to be intimately linked with the state of an excess O hole doped in the  $\text{CuO}_2$  plane. In the present paper, we investigate the nature of this single-hole state theoretically. Special attention is paid to the influences from the O sublattice. We argue that in the case of strong covalent bonding, the motions of O atoms can be very anisotropic: these atoms may find themselves in a single-well potential when moving perpendicular to the Cu–O–Cu bond directions, but in a double-well potential when moving along that direction. As a consequence, a two-plaquette localized polaron state can be realized for a doped hole in the  $\text{CuO}_2$  plane. The energy of this state is evaluated via a variational method. It breaks several symmetries locally and these symmetries have been experimentally observed to be broken also in the pseudogap phase.

(Some figures may appear in colour only in the online journal)

## 1. Introduction

Perovskite compounds and their variants have been the source of very rich physics, including ferroelectricity (FE) [1], quantum paraelectricity [2], multiferroicity [3] and high  $T_c$  superconductivity (HTC) [4], for example. The physically active ingredients are unanimously thought to be the  $\text{BO}_2$  planes, B being some transition metal element. These planes are usually interconnected through apical O atoms and 3D behavior results. When such interconnections are suppressed, 2D behavior could then take place. Despite the importance of B ions, O ions should also be essential partially because of their crystal positions: each O atom resides between two B atoms in line. For example, the spins on B ions in magnetic materials can be coupled mainly via the middle O ion [5] and the alignment of the polar modes in ferroelectrics based on B atoms requires the presence of O atoms [6]. In some HTC models, the O vibrations are even supposed to be the main pairing channel [7].

Even in the insulating phase of HTC, the O vibrations can play an important role. Let  $x$  and  $T$  stand for the concentration of doped holes and temperature, respectively. At low  $T$  and small  $x$ , these excess holes have been experimentally

demonstrated to be randomly quenched and localized [8]. Plenty of work has been done to address problems regarding the nature of this localized state. No consensus has been reached, but such a state is likely to be underlying the pseudogap phase and therefore may be of fundamental importance in understanding the entire phase diagram of HTC [9]. In an early work, an excess hole was thought to be confined exactly on an O site [10].

However, the above picture does not respect the underlying lattice symmetry and disregards the formation of Zhang–Rice singlets (ZRS) [11]. In [12], we forwarded an improvement, whereby a localized hole was proposed to simultaneously take up two adjacent Zhang–Rice states through tunneling. A Zhang–Rice state is mostly an orthonormal version of the symmetric combination of the four  $\sigma$ -type O 2p orbitals belonging to a  $\text{CuO}_4$  plaquette, namely  $\phi_i \sim \frac{1}{2} \sum_j (-1)^j p_{ij}$ , where  $i$  labels the plaquette and  $i_{j=1-4}$  indices the involved O sites [12]. A hole in such a state was shown to be strongly coupled to the Cu spin and results in a composite object called ZRS [11]. As shown in [12], a ZRS has to be somewhat deformed by tunneling effects. Consequently, a certain two-plaquette state was conceived for the hole and it was argued to be stable by virtue of

O-sublattice distortions. The required distortion seems to be supported by experiments [13] and computer simulations [14]. A very similar idea [15] has been put forward to interpret the checkerboard pattern detected in the pseudogap phase [16].

The primary goal of the present paper is to investigate further the nature and stability of the two-plaquette state. In the first part of this paper, we examine the on-site O potential surface,  $E_O$ . This is to be done on a relatively general ground and the results should therefore be valid not only for  $\text{CuO}_2$  but also many other  $\text{BO}_2$  planes. An interesting example concerns the  $\text{TiO}_2$  planes in barium titanate. For this compound, theoretical models [17, 18] often design anisotropic O vibrations: along the Ti–O–Ti direction a double-well potential is imposed whereas in the perpendicular direction a simple harmonic potential is imposed on O ions. Such anisotropy has also been implicitly adopted in some HTC models [19]. Nevertheless, the origin has remained unknown until now. In the second part, we show that the two-plaquette hole state can be highly robust in the presence of such an anisotropic O environment. It is shown to be essentially a mixture of ZRS and local O distortions and therefore a polaron state. Its formation energy is evaluated and thus its stability can be assessed. We note that such a state cannot be captured by the conventional t–J model. The failure of the latter was also noticed in, for example, [20].

## 2. Oxygen potential surface

The  $E_O$  is to be achieved by studying a local B–O–B cluster model [21], which includes the orbitals  $\text{O}_{2p}$  and  $\text{O}_{3s}$  in addition to an effective B  $d_{x^2-y^2}$ <sup>1</sup> with respect to the original lattice under no distortion. Notice that the crystallographic O position (equally distant from the B ions on the cluster) is highly symmetric. Usually, such positioning is unstable toward Jahn–Teller-type distortions. Displacing the O ion away from either B ion is likely to lift the lattice energy but at the same time result in lower electronic energy. Symmetry breaking often leads to extra electronic orbital mixing and hence lowers electronic energy due to the so-called ‘level repulsion’ effect. By off-center shifting the O ion, two consequences are immediate: (a)  $\text{O}_{2p}$ – $\text{O}_{3s}$  hybridization and hence O polarization; more importantly, (b) a change in the B–O covalent bonding energy. In what follows, we show that both (a) and (b) tend to lower the electronic energy and, in particular, because of (b) the  $E_O$  appears very anisotropic. The energy calculations will be based on the Born–Oppenheimer approximation.

We begin with the O polarization. Suppose the cluster lies along the  $x$  direction. Denote the displacements of the O atom by  $\vec{u}$ . The model Hamiltonian in accordance with

symmetry can be written as  $W_1 = \frac{\Delta_{sp}}{2}(\hat{N}_s - \sum_{\rho=x,y,z} \hat{N}_{p\rho}) + \lambda \sum_{\rho=x,y,z} (\hat{T}_{s\rho} + \hat{T}_{s\rho}^\dagger) u_\rho$ . Here  $\Delta_{sp}$  denotes the energy difference (that has already been renormalized by the crystal field) between the 3s and 2p orbitals,  $\hat{N}_{v=s,p_x,p_y,p_z} = |v\rangle\langle v|$  is the number operator and  $\hat{T}_{s\rho} = |s\rangle\langle p_\rho|$  is the operator that transfers an electron from the O 2p<sub>ρ</sub> to the O 3s. In addition, the  $\lambda$  is a real constant and measures the likelihood of electron transfer assisted by a local O phonon. Besides,  $\hat{T}^\dagger$  is the conjugate of  $\hat{T}$ . The O ion can be polarized through events represented by  $\hat{T}$  and  $\hat{T}^\dagger$ . This model can be easily solved to get the ground state energy, whose exact form is not needed here. We only write down the relevant part for small  $\vec{u}$ . Assuming  $\frac{\lambda u}{\Delta_{sp}}$  be a small quantity, it is

$$E_1 = -\frac{4\lambda^2}{\Delta_{sp}} \vec{u}^2. \quad (1)$$

In obtaining this expression we have counted the spin degeneracy. Note that this  $E_1$  is negative, implying that the electronic energy is lower for finite  $\vec{u}$ . Moreover,  $E_1$  does not depend on the direction of  $\vec{u}$  and therefore no anisotropy comes here.

Now we demonstrate that the covalent bonding between the B  $3d_{x^2-y^2}$  (referred to as  $|d_\pm\rangle$  below for the left and right orbital, respectively) and the O 2p<sub>x</sub> ( $|p\rangle$  below) (see footnote 1) orbital could alter the above picture and cause the anisotropy. A proper Hamiltonian to describe the B–O bonds involved in the cluster may be formed as  $W_2 = \frac{\Delta_{dp}}{2}(|d_+\rangle\langle d_+| + |d_-\rangle\langle d_-| - |p\rangle\langle p|) + [|p\rangle(t_+(d_+| + t_-(d_-|) + \text{h.c.})]$ . Here  $\Delta_{dp}$  denotes the energy mismatch between the d and p orbitals. The hybridization coefficients can be expressed as  $t_\pm = \delta_\pm \pm t_0$ , where the signs come from the phases of the d- and p-type wavefunctions, and the  $\pm t_0$  is the orbital overlap without phonons while  $\delta_\pm$  denotes the correction due to O phonons. By symmetry considerations, we in general have  $\delta_\pm = au_x \pm b\{u_y^2 + u_z^2\}$ , for small  $\vec{u}$ s. Here  $a$  and  $b$  are constants and for Coulombic electron–ion interactions, one roughly has  $a^2 \sim 2t_0b$ . The  $W_2$  can be analogously solved and the ground state energy can be obtained. Again we only write down the relevant part, which is

$$E_2 = \frac{8t_0b}{\Delta_{dp}}[u_y^2 + u_z^2] - \frac{4a^2}{\Delta_{dp}}u_x^2 \quad (2)$$

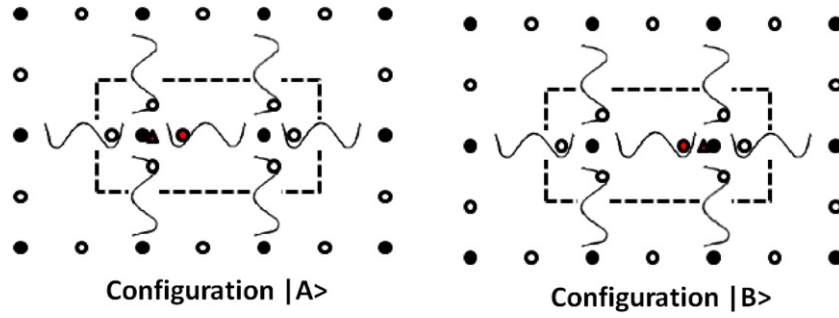
where the spin degeneracy has been counted. We see that  $E_2$  is anisotropic: positive for  $u_y$  and  $u_z$  but negative for  $u_x$ . The stronger the bond is, the more anisotropic this term will be.

To stabilize the crystal, one should take into account the lattice energy. In line with cubic symmetry, this energy may be generally written as  $\alpha_4 u_\rho^4 + \alpha_2 u_\rho^2$ , with  $\alpha_{2,4}$  being constants. Eventually, we write

$$E_O = \alpha_4 u_x^4 + \gamma u_x^2 + \sum_{x=y,z} [\alpha_4 u_x^4 + \bar{\gamma} u_x^2].$$

Now the vibration frequencies  $\omega_{l,t}$  (here  $l$  refers to the bond direction and  $t$  the perpendicular directions) can be evaluated, e.g. by the self-consistent harmonic approximation [22], in which one writes  $\frac{1}{2}m\omega_l^2 = \gamma + \alpha_4 \langle u_x^2 \rangle$  and similarly  $\frac{1}{2}m\omega_t^2 = \bar{\gamma} + \alpha_4 \langle u_x^2 \rangle$ . Here  $m$  denotes the oxygen mass and  $\langle \dots \rangle$  dictates

<sup>1</sup> When talking of the  $\text{Cu}^{2+}$  in cuprate superconductors, one must take into account the Hubbard repulsion  $U$ . In the case of large  $U$ , the  $d_{x^2-y^2}$  splits into upper and lower Hubbard orbitals. At half-filling, only the lower one is always fulfilled. In energy, the upper lies at the top, the lower at the bottom and the O 2p is in between. Therefore, the relevant orbitals entering equation (3) are essentially the O 2p<sub>x</sub> and the upper Hubbard orbital. Since the latter is empty and the corresponding  $\Delta_{dp}$  is positive, the form of  $E_2$  should still apply.



**Figure 1.** The polaron state as explained in the text. Solid circles denote the Cu sites, empty ones the O sites and the triangle the doped hole. A plaquette is made of a Cu site and the O sites surrounding it. The dashed rectangle encloses the polaron, which is supposed to be in a combination of the two low energy states  $|A\rangle$  and  $|B\rangle$ . Such states can be stable at low temperatures, as argued in the text.

the thermal average. In  $E_0$ ,  $\gamma = \alpha_2 - (\frac{4\lambda^2}{\Delta_{sp}} + \frac{4a^2}{\Delta_{dp}})$  and  $\bar{\gamma} = \alpha_2 + \frac{8t_0b}{\Delta_{dp}} - \frac{4\lambda^2}{\Delta_{sp}}$ .  $\omega_l < \omega_t$  ensues provided that  $\gamma < \bar{\gamma}$ . Thus, the  $l$  mode should dominate the O phonon sector at low energies.

### 3. The two-plaquette polaron state

In the rest of this paper we make connections with HTC, where only a single  $\text{CuO}_2$  sheet has to be considered. A profound mystery lies with the pseudogap phase, which initiates with a glassy-like state as a result of quenched holes [23]. It is characterized by a  $T^*(x)$  line on the lightly doped side in the  $x$ - $T$  phase diagram. Experimentally, the electronic excitations in this region were found extremely incoherent at small  $x$  prior to the onset of superconductivity and gradually gaining coherence at increasing  $x$  [9]. The incoherence is likely to originate from hole localization occurring at small  $x$ , as observed by STM probes [24]. The nature of this localized hole state was recently hypothesized to represent a compound object (a polaron) comprising a doped hole, local magnons and local lattice distortion [12]. In what follows we elaborate on this picture and give a more quantitative analysis.

The undoped  $\text{CuO}_2$  sheet is an antiferromagnet configured as  $\text{Cu } 3d^9 \text{ O } 2p^6$ . Each Cu site is furnished with a spin. Now we dope a very small number of holes into this sheet, which are supposed to be randomly quenched and populate Zhang–Rice states [12, 11]. Their correlations may be ignored since their mean distance is large at small  $x$ . We therefore focus on a single hole and imagine it falls on a Zhang–Rice state  $|1\rangle$ . What will happen to it? First, it will interact strongly with the Cu spin  $\vec{S}_1$  in the same plaquette, as first proved in [11]. Second, it will tunnel to another Zhang–Rice state  $|2\rangle$  in proximity to minimize its kinetic energy [12]. Third, this tunneling also allows it to interact with Cu spin  $\vec{S}_2$ . Fourth, it will pull the O sites around to maximize its magnetic interactions with Cu spins, since the Zhang–Rice coupling strength  $K$  is sensitive to the overlap between the  $d_{x^2-y^2}$  and  $p$  orbitals, which in turn is a function of the Cu–O bond length. As reasoned in [12], the localized state should be primarily determined by the interplay between the Zhang–Rice states  $|i; i = 1, 2\rangle$ , the Cu spins  $\vec{S}_{1,2}$  and the O lattice distortion.

The situation has been sketched in figure 1, where the dashed rectangle encloses the main part of the polaron of interest.

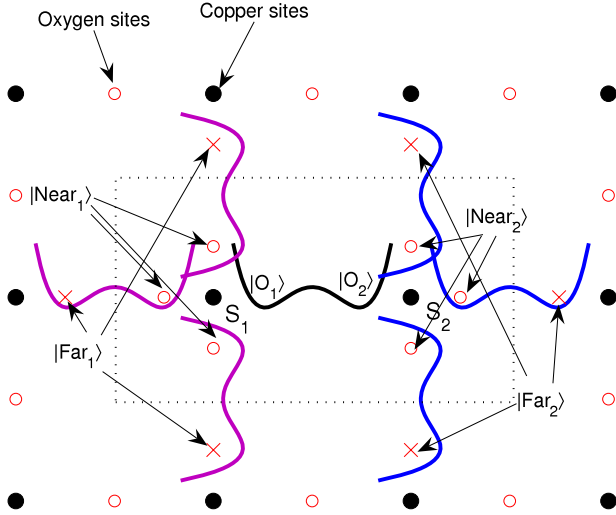
#### 3.1. The model of oxygen vibrations

To make progress, we postulate the validity of  $E_0$ . This will considerably simplify the analysis. According to  $E_0$ , we may safely neglect the O vibrations perpendicular to the Cu–O bonds, since they are energetically unfavorable. As seen in figure 2, there is an O atom (whose displacement relative to its central position along the bond will be denoted by  $x$ ) that is shared by both plaquettes. Among the remaining O atoms, the left three form an aggregate to be referred to as  $A_1$  while the right three form an aggregate  $A_2$ . For each aggregate, when restricted to bond-direction vibrations, only three eigenmodes (stemming from the displacements of the three O atoms in the aggregate) can exist and evidently among them the one that couples most efficiently to the doped hole and the Cu spins should have all the constituting O atoms moving in phase. Let us denote the coordinate of this mode by  $x_{i=1,2}$ , where  $i = 1$  and  $i = 2$  correspond to  $A_1$  and  $A_2$ , respectively. All other modes are irrelevant and then omitted hereafter.

To model the O vibrations, we assume a negative  $\gamma$  in  $E_0$ . This should be possible if the covalent bond and the O polarizability are strong enough. Under this assumption, one ends up with a double-well potential for  $x$  and  $x_{1,2}$ . We may further reduce the O phonon Hilbert space by including only states in the vicinity of the potential minima. For the shared O atom, the states corresponding to the left and right potential minima (see figure 2) will be denoted by  $|O_1\rangle$  and  $|O_2\rangle$ , respectively. For the aggregate  $A_i$ , we use  $|\text{Far}, i\rangle$  for the state corresponding to the minimum that is far from the  $i$ th Cu site while  $|\text{Near}, i\rangle$  for the one closer to this Cu site. Note that each of these states can approximately be viewed as the ground state of a harmonic oscillator that mimics the low energy O vibrations about the corresponding minimum. We now see that the low energy sector of the subsystem of O phonons is equivalent to a system of three pseudo-spins.

The Hamiltonian for these pseudo-spins can in general be couched as

$$\begin{aligned}
 H_O &= \Omega(\tau_1^x + \tau_2^x) + \Omega' \tau_0^x, \\
 \tau_i^x &= |\text{Far}, i\rangle\langle\text{Near}, i| + |\text{Near}, i\rangle\langle\text{Far}, i| \\
 \tau_0^x &= |O_1\rangle\langle O_2| + |O_2\rangle\langle O_1|
 \end{aligned}$$



**Figure 2.** Illustration of the states introduced in the text. The curves stand for the double-well potential surfaces  $E_O$  for O atoms. The dashed rectangle indicates the polaron region.

where  $\tau^x$  is the second Pauli matrix that flips the pseudo-spins (i.e. tunnels between the two potential minima) and  $\Omega$  and  $\Omega'$  represent the flip energies that are conversely dependent on the minima separation  $2l_0 = \sqrt{\frac{2|\gamma|}{\alpha}}$ . We expect  $\Omega' \gg \Omega$ , because the aggregates are much heavier than a single O atom. The eigenstates of  $H_O$  are then given by direct products of the eigenvectors of  $\tau_{0,1,2}^x$ , namely

$$|s_0, s_1, s_2\rangle = |s_0\rangle \otimes |s_1\rangle \otimes |s_2\rangle,$$

where  $s_{0,1,2} = \pm 1$  are phase factors and  $|s_0\rangle = \frac{1}{\sqrt{2}}(|O_1\rangle + s_0|O_2\rangle)$  as well as  $|s_i\rangle = \frac{1}{\sqrt{2}}(|\text{Far}, i\rangle + s_i|\text{Near}, i\rangle)$ . Note that  $x$  and  $x_i$  have vanishing averages over these states.

### 3.2. Variational state for the polaron

We proceed to conceive an approximate ground state for a doped hole according to the analysis prescribed above. Let us for the moment forget about the kinetic energy (i.e. tunneling effects) of the hole and the O atoms. We can then let the hole simply settle on, say, the left plaquette. The ground state would then be basically the Zhang–Rice singlet (denoted by  $|\text{ZRS1}\rangle = \frac{1}{\sqrt{2}}[|1 \uparrow; S_{1z} = \downarrow\rangle + |1 \downarrow; S_{1z} = \uparrow\rangle]$ <sup>2</sup> with all O atoms pulled into the potential wells closer to the Cu site, namely  $|\text{ZRS1}\rangle \otimes |\text{Near}, 1\rangle \otimes |O_1\rangle$ . Now we relax the O atoms and render it tunnel. The O states are then expected to be admixed and the ground state needs to be modified. By intuition, we may write it as

$$|A\rangle = |\text{ZRS1}\rangle \otimes |\lambda; 1\rangle \otimes |\text{Near}_1\rangle \otimes |\text{Near}_2\rangle$$

<sup>2</sup> To be more complete, we should also specify the value of  $S_{2z}$  when writing down  $|\text{ZRS1}\rangle$ , namely we should put  $|\text{ZRS1}; S_{2z}\rangle$ . Here  $S_{2z}$  can be either up or down. Similarly, instead of  $|\text{ZRS2}\rangle$ , we should put  $|\text{ZRS2}; S_{1z}\rangle$ . As shown in [12], to allow ZRS hopping, one should require  $S_{2z}$  and  $S_{1z}$  take the same value (we choose  $\uparrow$  for definiteness). Therefore, that  $|\text{ZRS1}\rangle \equiv |\text{ZRS1}; \uparrow\rangle$  and  $|\text{ZRS2}\rangle \equiv |\text{ZRS2}; \uparrow\rangle$  should be understood throughout the rest of the paper.

where

$$|\lambda; 1\rangle = \frac{1}{\sqrt{1+\lambda^2}}(|O_1\rangle - \lambda|O_2\rangle),$$

with  $\lambda$  being a parameter to be determined via variational methods. Similarly, if the hole settles on the right plaquette, the state should be

$$|B\rangle = |\text{ZRS2}\rangle \otimes |\lambda; 2\rangle \otimes |\text{Near}_1\rangle \otimes |\text{Near}_2\rangle$$

with

$$|\lambda; 2\rangle = \frac{1}{\sqrt{1+\lambda^2}}(\lambda|O_1\rangle - |O_2\rangle).$$

Finally, we allow the hole to tunnel at a certain rate. As shown in [12], this will mix  $|A\rangle$  and  $|B\rangle$  to yield the desired polaron state:

$$|P\rangle = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle),$$

which is an equi-weighted combination of  $|A\rangle$  and  $|B\rangle$ , as required by the symmetry between the left and right plaquettes. For comparison, we also write down the normal state as

$$|N\rangle = \frac{1}{\sqrt{2}}(|\text{ZRS1}\rangle + |\text{ZRS2}\rangle) \\ \otimes |s_0 = -1, s_1 = -1, s_2 = -1\rangle$$

which is found by taking  $\lambda = 1$  in  $|P\rangle$ . The polaron formation energy  $E_f$  is hence given by the difference between the energies of  $|P\rangle$  and  $|N\rangle$ . In writing down  $|P\rangle$  and  $|A\rangle$  as well as  $|B\rangle$ , we have assumed that the aggregates always stay in their respective  $|\text{Near}\rangle$  states. This is realistic considering that  $\Omega$  is small and that the ZRS energy can be lowered a lot if the aggregates are pulled closer to the Cu spins.

### 3.3. The polaron model

To evaluate  $E_f$ , we need to build a Hamiltonian to model the polaron depicted in figures 1 and 2. It is convenient to partition the Hamiltonian into two parts, the first simply containing terms within the polaron while the second describes the interactions of the polaron with the rest of the system. In accord with the scenario established above, the minimum scheme concerning the first part must encapsulate the hopping (with energy  $-t_0$ ) of the hole between the two Zhang–Rice states  $|i\rangle$  involved in the polaron, the local Zhang–Rice coupling between the hole's spin  $\vec{s}$  and the left and right Cu spins  $\vec{S}_{1,2}$  (with strengths  $K(x_{1,2}, x)$ ) and the coupling between  $\vec{S}_1$  and  $\vec{S}_2$  (with strength  $J'$ ). Presuming localization of the hole and distortion of O aggregates, we may refrain from the hole hopping out of the polaron, i.e. hopping out of the rectangle is frustrated. Thus, the second part  $H'$  should contain only nearest-neighbor interactions (with strength  $J(x_i)$ ) between the  $\vec{S}_{1,2}$  and the Cu spins outside the polaron. As suggested in their forms, we have allowed the  $K(x_i, x)$  and  $J(x_i)$  to depend on O positions and hence spin-phonon interactions have been included, which entails



the inclusion of  $H_O$  for a complete description. Summarizing, we write the Hamiltonian as

$$H = J' \vec{S}_1 \cdot \vec{S}_2 + \vec{s} \cdot [K(x_1, x) \vec{S}_1 + K(x_2, x) \vec{S}_2] - t_0(|1\rangle\langle 2| + \text{h.c.}) + H_O + H' \quad (3)$$

where

$$H' = \sum_{\langle i,j \rangle, i=1,2} J(x_i) \vec{S}_i \cdot \vec{S}_j,$$

with the pair  $\langle 1, 2 \rangle$  excluded in the summation. In the presence of antiferromagnetic order, one may write  $H' \approx \sum_{i=1,2} J(x_i) \vec{S}_i^z \cdot (-1)^i M_0$ , where  $M_0$  is the average magnetization. For small  $x_i$  and  $x$ , we may expand  $J(x_i)$  and  $K_{x_i, x}$  as  $J(x_i) = J_0 - x_i^2 \frac{\partial^2 J}{\partial x_i^2}$  and  $K(x_i, x) = K_0 + (3x_i + (-1)^i x) \frac{\partial K}{\partial x_i}$ , where all derivatives are taken at  $x = 0$  and  $x_i = 0$  and the factor 3 comes from the fact that an aggregate consists of three O atoms; the linear term in  $J(x_i)$  vanishes by virtue of symmetry and that  $\frac{\partial K}{\partial x_i} = \frac{\partial K}{\partial x} > 0$  should hold. Experimentally, it is known that Zhang–Rice singlets are very stable, indicating the dominance of the  $K$ -leading terms over others.

### 3.4. Polaron formation energy and its stability

Now we evaluate  $E_f$  and assess the stability of the polaron state. By definition, we have  $E_f = \langle P|H|P \rangle - \langle N|H|N \rangle$ . It is useful to notice that both  $H'$  and the  $J'$ -leading terms vanish when averaging over  $|P\rangle$  and  $|N\rangle$ , since they vanish over  $|ZRS\rangle$ . Making use of the fact that  $x = l_0(|O_2\rangle\langle O_2| - |O_1\rangle\langle O_1|)$  and that  $x_i = l_0(|\text{Near}_i\rangle\langle \text{Near}_i| - |\text{Far}_i\rangle\langle \text{Far}_i|)$  and after some algebra, we find

$$E_f = \Omega + \left( \Omega' + \frac{t_0}{2} \right) \frac{(1 - \lambda)^2}{1 + \lambda^2} - \frac{3g}{4} \frac{1 - \lambda^2}{1 + \lambda^2} - \frac{9g}{4} \quad (4)$$

where  $g = l_0 \frac{\partial K}{\partial x}$ . The first term in  $E_f$  is no more than the lattice energy cost of pulling the aggregates while the last one is the energy lowering of ZRS due to that pulling. The energy cost as expressed in the second term has two contributions: the increasing of the lattice energy of off-center shifting the shared O atom and the increasing of the hole's kinetic energy when the hopping is impeded by that off-center shift. Meanwhile this shift lowers the ZRS energy by the amount contained in the third term. Minimizing this expression, we arrive at

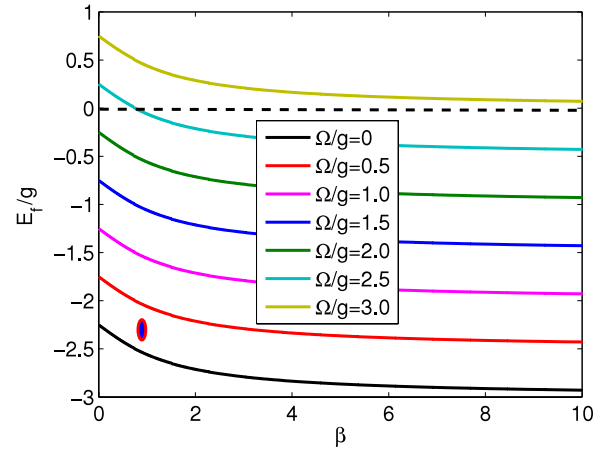
$$\lambda = \sqrt{1 + \beta^2} - \beta, \quad \beta = \frac{3g}{4\Omega' + 2t_0}.$$

Substituting it back,  $E_f$  can be found. When  $\beta \ll 1$ , one has  $\lambda \approx 1 - \beta$ . In this case, the stability condition that  $E_f < 0$  translates into the following:

$$\frac{\Omega}{g} < \frac{9}{4} \quad (5)$$

which is independent of  $\beta$ , while if  $\beta \gg 1$ , one has  $\lambda \approx \frac{1}{2\beta}$  and the stability condition becomes

$$\frac{\Omega}{g} < 3. \quad (6)$$



**Figure 3.** The polaron formation energy  $E_f$  as a function of  $\beta$  for various  $\frac{\Omega}{g}$ . The spot corresponds to an estimate given in the text.

For general values of  $\beta$ , we have plotted  $E_f$  versus  $\beta$  in figure 3, where it is exhibited that, for  $\Omega/g < 3$ , that  $E_f < 0$  is much likely to be gratified and the polaron picture is quite plausible. In addition, it is seen that the main energy to be circumvented is that of pulling the aggregates while the off-center shift always results in a lower total energy.

It is possible to make a realistic estimate. Notice that expression (4) involves four parameters:  $t_0$ ,  $\Omega'$ ,  $\Omega$  and  $g$ . Let us look at  $t_0$  and  $g$  first. From the expression of  $K$  as given in [11, 12], we learn that  $K \sim V^2[\frac{1}{\Delta} + \frac{1}{U-\Delta}]$ , where  $V$  is the overlap between the Cu  $d_{x^2-y^2}$  and O p orbital wavefunctions and  $U$  is the Hubbard repulsion on Cu sites whereas  $\Delta$  is the charge transfer gap. Assuming an exponential dependence of  $V$  on the Cu–O bond length variation  $\delta$ , i.e.  $V \approx V_0 \exp(\frac{\delta}{y_0})$ , we obtain  $g \approx V_0^2[\frac{1}{\Delta} + \frac{1}{U-\Delta}]^2 \frac{l_0}{y_0}$ . Taking  $V_0 \sim 1.3$  eV,  $U \sim 10.5$  eV,  $\Delta = 3.6$  eV and  $\frac{l_0}{y_0} \sim 0.1$  from [25], we would get  $g \approx 100$  meV. Here  $y_0$  denotes the average Cu–O bond length. Similarly, we have  $t_0 \sim V_0^2[\frac{1}{\Delta} + \frac{1}{U-\Delta}] \sim 0.2$  eV. On the other hand,  $\Omega'$  and  $\Omega$  should be much smaller, since they are lattice vibration energies. Actually, in perovskite compounds they are often found to be less than  $200 \text{ cm}^{-1}$ , amounting to about 20 meV. Given such estimates, we obtain  $\beta \sim 0.6$ ,  $\frac{\Omega}{g} \sim 0.2$  and  $E_f \approx -2g \sim -t_0$ , which is a very large portion of energy. Although very rough, the estimate, which is indicated by the spot in figure 3, shows that the polaron state can be very stable.

## 4. Discussion

A direct corollary of the two-plaquette picture are the spatial Cu–O bond length variations at low doping. According to this picture, a doped hole will form a polaron with two Cu spins by attaching the O atoms nearby. As a result, three pairs of anomalous Cu–O bonds will be created locally. With  $x$  being the doping level, we then anticipate that the fraction of these anomalous bonds amounts to about  $1.5x$ . In [13], experiments aimed at detecting local lattice distortions were performed on  $\text{La}_{0.75}\text{Sr}_{0.15}\text{CuO}_4$ . The authors indeed found elongated Cu–O bonds, whose fraction was about 20% and close to the expected value. Their experiments seemingly did

not detect the shorter bonds, but this may be due to limited resolution.

Locally, the electronic density distribution associated with the polaron state should be bond-centered and look like a dipole, with the maxima occurring on the ends, thereby breaking the fourfold rotational symmetry. Interestingly, this type of symmetry breaking has already been discovered experimentally [26, 27]. In [27], the dipole pattern seems clearly detected in the pseudogap phase and the onset temperature of the latter seems to be tracking the pattern forming temperature.

The as-constructed polaron is magnetic in the sense that it carries a small but finite magnetic moment, which has been demonstrated in [12]. Largely, it is due to an unusual double-exchange effect: the hole couples simultaneously via hopping to both Cu spins and therefore acts to align the spins, thence generating finite magnetism and breaking time-reversal symmetry. The moment for typical material parameters could be easily as large as  $0.1 \mu_B$  [12]. This should be apparent from, for example, Kerr rotation and neutron scattering. Experiments have indeed observed unusual magnetic phenomena that are most pronounced in the pseudogap phase of very underdoped materials [28, 29].

In view of all these appealing features, it is appropriate to speculate on the connections of the polaron state with the pseudogap phase. As already said, they seem to have the same symmetries broken locally and the pseudogap onset seemingly synchronizes with the appearance of these features. More interesting evidence is discussed below.

Recently, simultaneous  $r$ - and  $k$ -space imaging techniques have revealed remarkable evolutions in the nature of the electronic excitations when the doping tends towards zero [9]. Such evolutions seem to place the polaron state (maybe its variants) as a strong candidate for the pseudogap state. As is well known, in underdoped samples two distinctive categories of electronic excitations can be seen at low energies and high energies, respectively. Utilizing the imaging technique, Kohsaka *et al* observed two classes of excitations: one is well defined in  $k$  space whereas the other is localized in  $r$  space [9]. The  $k$ -space excitations were proved to be simply the Bogoliubov quasi-particles generated out of the superconducting phase. On the other hand, the  $r$ -space excitations were found with higher energies and were found to destroy the aforementioned symmetries locally. They demonstrated that these  $r$ -space excitations are just the pseudogap excitations. Intriguingly, upon decreasing doping the phase volume occupied by the  $k$ -space excitations shrinks rapidly and the spectral weight is transferred to the  $r$ -space states. In light of our reasoning, one may conjecture that these  $r$ -space states could be intimately linked with the polaron states described above. If so, one should have  $|E_f| \sim T^*$ .

In the present work, we have focused on the single-polaron aspects. Future work is needed to investigate the

inter-polaron correlations (which should be of importance to superconductivity) and the inter-relations between polarons and the antiferromagnetic background.

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